GraphML Assignment 2 - María Sánchez Domínguez

Spatial Organization Report

Overview

This report interprets the results of a graph-based analysis applied to an architectural floor plan. The building is modeled as a spatial graph where rooms and spaces are represented as nodes, and adjacencies or connections between them are represented as edges.

Five metrics were computed and visualized: the base **Centrality Graph** (spatial network), **Closeness Centrality**, **Degree Centrality**, **Community Detection**, and **Shortest Paths**.

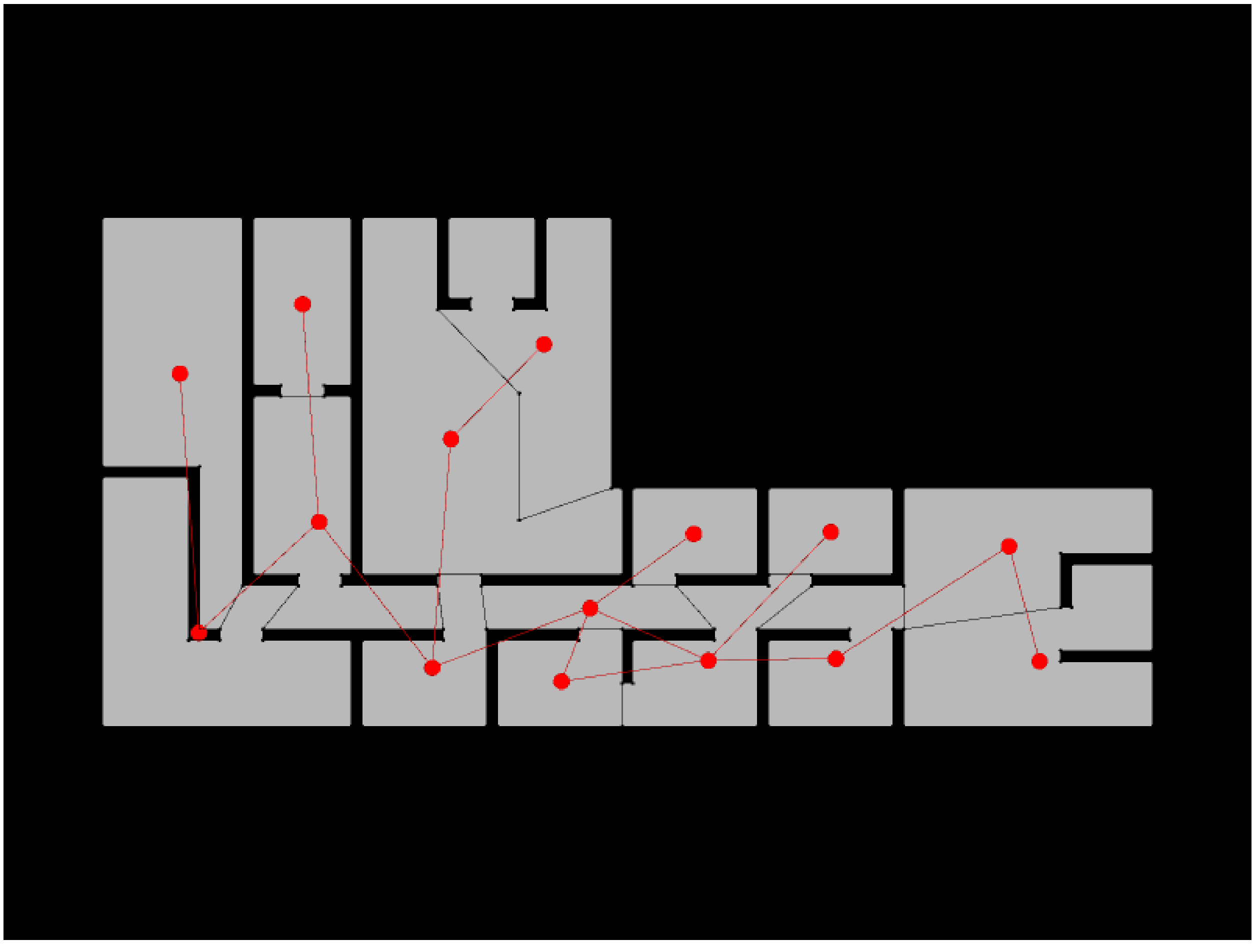
The floor plan reveals a building with two distinct morphological zones: a compact, multi-room **left wing** with vertically stacked spaces, and a long, predominantly horizontal **right corridor zone** with smaller rooms branching off it. This overall shape strongly influences every metric computed.

A Note on Grid Resolution

It should be noted that, due to unknown driver incompatibility or otherwise unresolved technical issues, the grid used as the basis for all analyses in this assignment was limited to a **2×2 resolution**. This significantly constrains the granularity of the results: a coarser grid produces fewer nodes, which in turn reduces the precision of all computed metrics and the detail of the visualizations.

The spatial patterns identified in this report — centrality hotspots, community boundaries, shortest path routes — should therefore be understood as approximate tendencies rather than fine-grained descriptions. Under ideal conditions, a higher-resolution grid would yield a richer graph with more nodes per space, enabling more nuanced readings of connectivity, hierarchy, and circulation.

1. Spatial Network — Centrality Graph



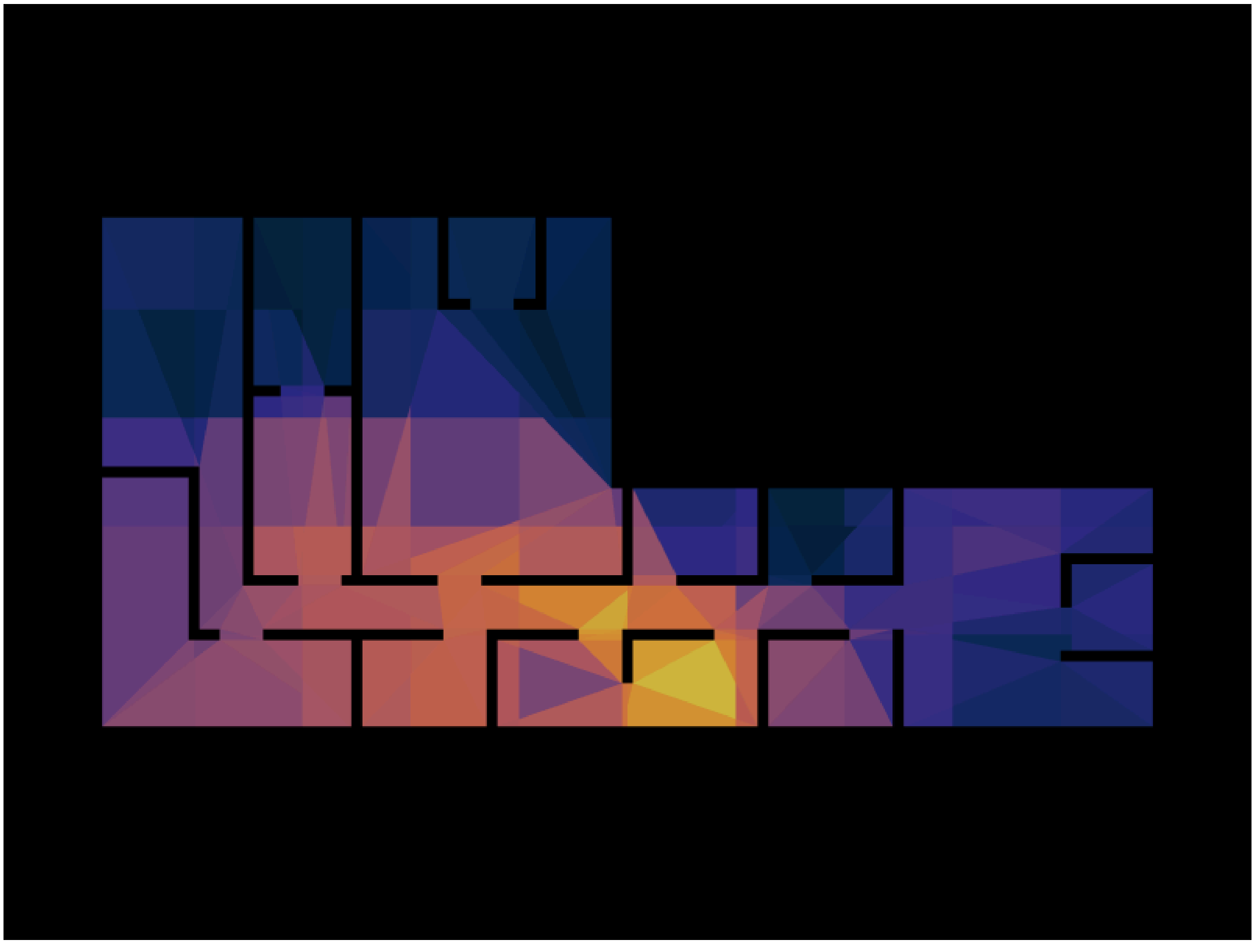
The centrality graph overlays the spatial network directly onto the floor plan. Each red dot represents a space (node) and each red line represents a direct spatial connection (edge) between two adjacent spaces.

The graph reveals a clear **bifurcation in connectivity style**:

- The **left wing** has a more branching, tree-like topology. Nodes here have fewer connections, and several spaces appear to be terminal — reachable only through a single parent node. This is typical of private or specialized rooms (offices, labs, studios) where direct access from a main corridor is the norm.
- The **right corridor zone** is essentially a **linear chain**, with nodes connected sequentially along the horizontal axis. A few cross-connections exist, but the dominant structure is serial — one must traverse intermediate spaces to reach the far end.

The most visually prominent hub in the network lies at the **junction between the left wing and the right corridor**, roughly center-left on the plan. This node sits at the convergence of multiple edges and acts as the primary distributional point for the entire building.

2. Degree Centrality



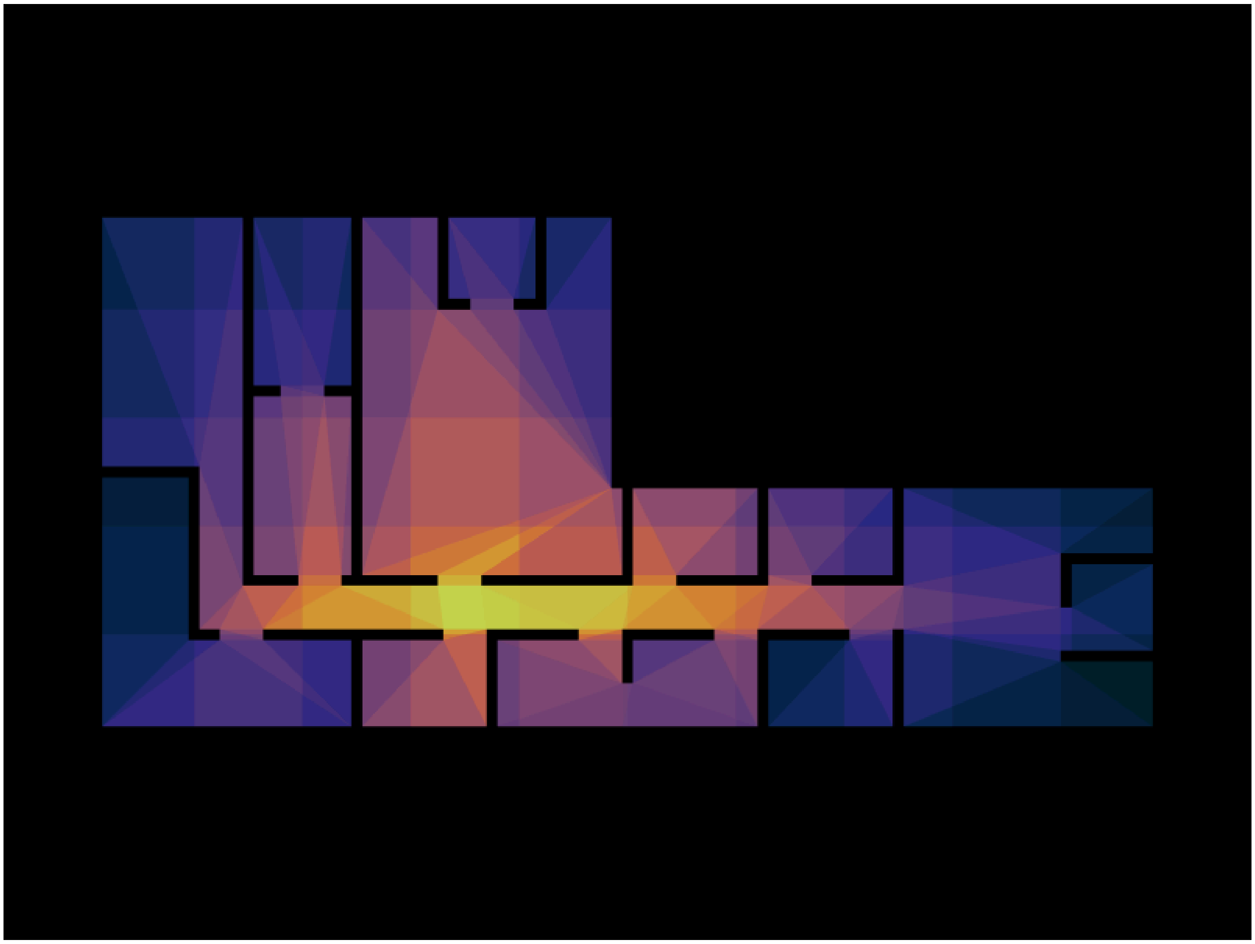
Degree centrality measures how many direct connections a space has. Spaces with high degree centrality (warm colors — yellow, orange) are directly adjacent to many other spaces; low-degree spaces (cool colors — dark blue, teal) are more isolated.

Key findings:

- The **highest degree centrality** is concentrated at the junction node between the left wing and the right corridor. This space connects upward into the vertical cluster of rooms, leftward to a peripheral room, and rightward into the corridor chain — making it the most immediately connected node in the network.
- The **left upper wing** shows moderate-to-low degree values (purple and mid-blue tones), indicating that spaces there are more specialized and less multiply connected. They tend to branch off a single vertical spine rather than connecting to several neighbors at once.
- The **far right end** of the corridor displays low degree centrality (dark blue), consistent with terminal spaces that connect back only through the corridor chain.

In architectural terms, the high-degree node is a **distributional hub** — likely a lobby, foyer, or main circulation point. Its removal would significantly impair access to both wings.

3. Closeness Centrality



Closeness centrality measures how quickly a space can reach all other spaces in the network. A space with high closeness centrality (bright yellow) minimizes the average travel distance to every other node; low closeness (dark blue) indicates peripheral or poorly connected spaces.

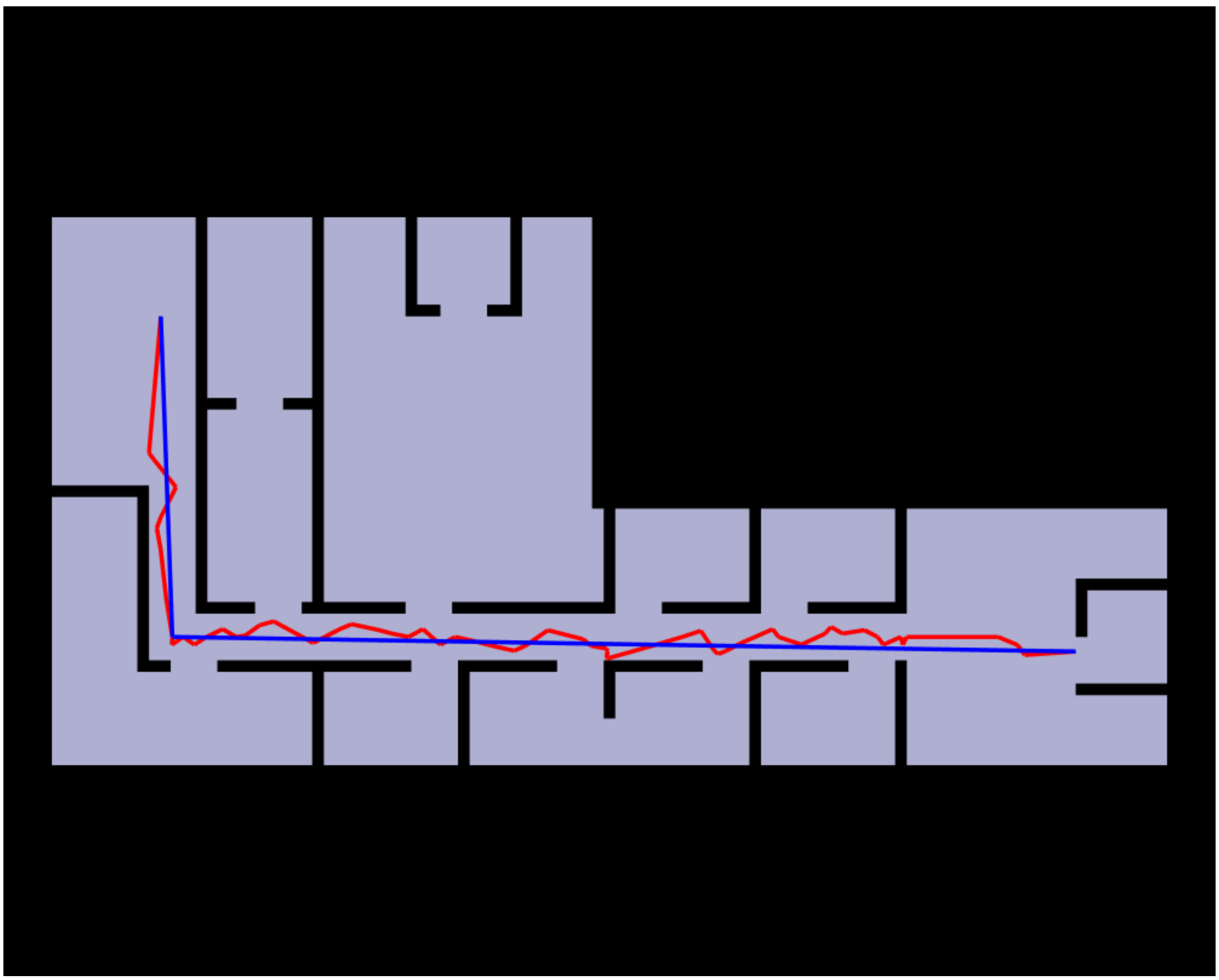
Key findings:

- The **brightest region** — highest closeness — sits near the center of the building, at the junction between the left wing and the horizontal corridor. This confirms what degree centrality suggested: this space is not only directly well-connected, but also optimally positioned relative to the entire network. It is the topological "center of mass" of the building.
- A **secondary warm region** (orange) extends into the upper portion of the left wing, suggesting that despite its branching structure, the vertical cluster of rooms is still reasonably close to the core.

- The **far left peripheral room** and the **far right end of the corridor** are the darkest (lowest closeness), meaning they are the most spatially remote spaces in the building. They require the most hops or the longest paths to reach any other part of the floor plan.

Architecturally, this metric highlights **accessibility imbalance**: a visitor or occupant at one end of the building faces a significantly longer journey to reach spaces at the other end. The central hub mediates this imbalance — but only imperfectly, since the corridor remains essentially linear.

4. Shortest Paths



This visualization shows two computed shortest paths (blue and red) overlaid on the floor plan. Both paths originate in the **upper-left wing** and terminate at the **far right end of the corridor**, tracing the minimum-cost route across the full extent of the building.

Key findings:

- Both paths converge almost immediately to the **central junction node**, confirming that there is effectively only one viable route connecting the left wing to the right corridor zone. This junction is a **mandatory waypoint** — a spatial bottleneck — for any trip between the two halves of the building.
- Once in the corridor, both paths run nearly parallel and horizontally to the right. The **red path** shows slight oscillations (likely following graph edges that zigzag slightly between corridor sub-spaces), while the **blue path** is smoother and more direct, possibly representing the Euclidean or topologically simplified route.
- The convergence of both paths through the same central node, and their alignment through the corridor, reinforces that the **main corridor is the primary — and perhaps sole — circulation spine** of the building's right zone.

The bottleneck quality of the central junction is worth noting: any disruption to that space (a closed door, a rearrangement, a fire) would sever the building's two halves from one another.

5. Community Detection

Community detection groups spaces that are more densely interconnected with each other than with the rest of the network. Each color represents a distinct cluster or zone.

Key findings:

The algorithm identifies approximately **six distinct communities**, which map coherently onto the floor plan's physical layout:

- **Dark navy (upper-left):** A cluster of rooms in the top portion of the left wing. These spaces share direct connections but have limited links outward, forming a self-contained sub-zone — likely a suite of related rooms (classrooms, offices, or labs).
- **Dark teal (lower-left):** The lower portion of the left wing, including what appears to be a large open space. Despite proximity to the dark navy cluster, it forms a separate community, suggesting a functional or spatial boundary between them (possibly a change in room type or access level).
- **Purple (center-left lower):** A transitional zone that bridges the left wing to the corridor. This community includes the high-centrality junction node, affirming its role as the connective tissue of the layout.
- **Pink/mauve (upper center):** A distinct community in the upper-center of the plan, corresponding to rooms that connect more to the upper vertical cluster than to the corridor — suggesting they are accessed primarily from the left wing's circulation.
- **Orange and yellow-green (center-right):** Two smaller communities in the mid-to-right portion of the plan. These may correspond to functionally distinct room types — for example, service spaces, secondary offices, or shared amenities — that are grouped together by the corridor but have internal sub-groupings.
- **Gold/orange (far right):** The rightmost zone forms its own community, reflecting the relative isolation of the building's far end from the central hub.

The community structure broadly maps onto a **three-zone functional model**: a private/specialized left wing, a transitional central hub, and a more distributed right corridor zone. This tripartite organization is a common pattern in institutional buildings.

Summary: Spatial Organization

Circulation Patterns

The building's circulation is **centralized through a single bottleneck junction** at the meeting point of the left wing and the corridor. All long-distance movement passes through this node. The right zone relies entirely on a **linear corridor** as its circulation spine, with no alternative routes — movement is serial and one-directional.

Hierarchy of Spaces

There is a clear spatial hierarchy. At the top sits the **central junction** (highest degree and closeness centrality), which governs access to both wings. Below it are the **corridor spaces**, which form the secondary tier — accessible but peripheral. At the base are **terminal rooms** at the far left and far right, which are the least connected and least accessible nodes in the network.

Accessibility and Connectivity

Accessibility is **unequally distributed**. The central hub and adjacent spaces enjoy excellent connectivity; peripheral spaces — especially the far-right corridor end and the isolated left-wing room — require traversal through multiple intermediary spaces. The building would benefit from additional connectivity between the left wing's upper cluster and the corridor, which could reduce travel distances significantly.

Functional Zoning

Community detection suggests the building is organized into at least three functional zones: a **private cluster** (left wing upper rooms), a **transitional/public zone** (central junction and surrounding spaces), and a **distributed service or shared zone** (right corridor). This aligns with typical institutional planning logic, where private or specialized spaces are grouped away from the main circulation and public or shared spaces anchor the core.